RAG EVAL 2.0 – Multimodal RAG Pipeline (High-Level Architecture)

1. Ingestion Pipeline

Process and index multimodal documents

Document Upload

- PDF
- DOCX
- PPTX
- MD
- Images

Document Parsing and Content Extraction

- Text extraction
- Table extraction + natural-language descriptions
- Image extraction (OCR/metadata)
- Diagram extraction (e.g., Mermaid → text)
- Preserve structure and provenance

Chunking and Enrichment

- Create semantic chunks by content type
- Add natural-language descriptions for tables and diagrams
- Assign metadata (content type, source, page, element ID)
- Generate embeddings for each chunk

Indexing

Vector Index (ChromaDB)

- Store chunk embeddings
- Store metadata (content type, source, etc.)

Keyword Index (MySQL)

- Store chunk text
- Enable keyword search (BM25) with metadata

Data Stores

ChromaDB (Vector Index)
Embeddings + Metadata

MySQL (Keyword Index)
Chunk Text + Metadata

2. Query & Answering Pipeline

Retrieve, rerank, and generate grounded answers using multimodal evidence

User Question (Browser or CLI)

Natural-language question about multimodal content (text, tables, images, diagrams, etc.)

Retrieval (Select Method)

- **Vector Search** (semantic similarity)
- **Keyword Search** (BM25)
- **Hybrid Search (RRF)** (combine vector + keyword)

Initial Candidate Retrieval

Retrieve expanded set of candidate chunks (e.g., top_k × 5 or minimum 20) with metadata

Content-Aware Reranking

Rule-based reranking using:

- Question intent (table, image, diagram)
- Evidence overlap (term matching)
- Content type metadata
- Original retrieval rank

Select final Top K (e.g., 3 chunks)

Grounded Answer Generation

Send question and top K multimodal chunks to Gemini

- Text, tables, images, and diagrams
- Generate grounded answer
- Include source citations

Response to User

Display answer with source references and relevant multimodal evidence (text, tables, images, diagrams, etc.)

Document Content Types Supported

- **Text** (plain text, paragraphs)
- **Tables** (native or extracted, with natural-language descriptions)
- **Images** (embedded or standalone, with OCR/metadata)
- **Diagrams** (e.g., Mermaid → natural-language relationships)
- **Slides & Documents** (PDF, DOCX, PPTX, MD, etc.) with mixed content

Retrieval Methods

- **Vector Search** (semantic similarity)
- **Keyword Search** (BM25)
- **Hybrid Search (RRF)** (reciprocal rank fusion)